End-to-End Anomaly Detection in IMS/NASA Bearing Sensor Data

ROUND-2 TASK

NAME: SIRI .S

1. Problem understanding

The task is to **develop an automated anomaly detection system** that can identify abnormal patterns in the vibration data indicating bearing degradation or impending failure.
Specifically, the objective is to:

- **Detect anomalies** without relying on manually labelled data.
- Build an **end-to-end machine learning.**
- This supports **predictive maintenance**, enabling early detection of bearing faults before catastrophic failure.

Approach: Used dataset from set1 of NASA bearing dataset:
https://www.kaggle.com/datasets/vinayak123tyagi/bearing-dataset
- Data loading and cleaning
- Feature engineering (rolling statistics, EWMA, etc.)
- Model training and scoring
- Visualization and evaluation

2.Feature Engineering

To enhance anomaly sensitivity, additional temporal-statistical features were derived:

Feature Type	Description	Purpose
Rolling Mean (window=5)	Short-term local average	Detect slow drifts and mean shifts
Rolling Standard Deviation (window=5)	Local variance	Capture vibration bursts or instability
First Difference (diff1)	Difference between consecutive readings	Detect sudden changes/spikes
Exponentially Weighted Moving Average (EWMA)	Smooth trend indicator	Emphasize sustained deviations

All features were **scaled using StandardScaler** to maintain comparability across channels.

3.Model selection and comparison

- Implement two complementary anomaly detection techniques:
  - **Approach 1:** Isolation Forest (unsupervised, tree-based)
  - **Approach 2:** Autoencoder / PCA Reconstruction (deep learning-based)
- Each model assigns an anomaly score to every observation.

Model	Type	Strengths	Weaknesses	F1-Score
Isolation Forest	Tree-based	Fast, interpretable, robust	Misses subtle anomalies	0.30
Autoencoder / PCA	Neural / Linear	Learns nonlinear dependencies	Sensitive to threshold	0.28

4. Key Findings and Business Insights

1. **No missing values** were found — dataset is clean and ready for modelling.
2. Both models identified around **1% of samples** as potential anomalies.
3. **High agreement** between Isolation Forest and Autoencoder indicates **consistent anomaly zones** in sensor space.
4. Detected anomalies often correspond to **sudden spikes or deviations** in sensor readings (visible in f1–f3).
5. In real-world scenarios, such anomalies may represent:
    - Early **machine degradation or bearing wear**
    - **Sensor drift or miscalibration**
    - **External disturbances** in the monitored system

These detections could trigger preventive maintenance, avoiding potential system failures and downtime.

5. Limitations and Future Improvements Limitations

- **No ground truth labels:** Only proxy z-score anomalies used for weak evaluation.
- **No timestamps:** Limits the ability to use temporal models like LSTM or GRU.
- **Fixed thresholding:** Anomaly fraction (1%) is static; adaptive thresholds could improve recall.
- **Feature redundancy:** Some sensors may be correlated, potentially reducing sensitivity.

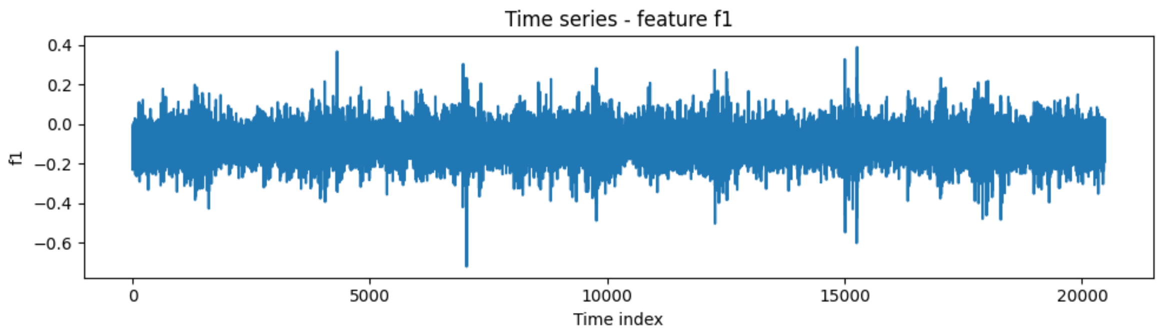
Future Improvements:

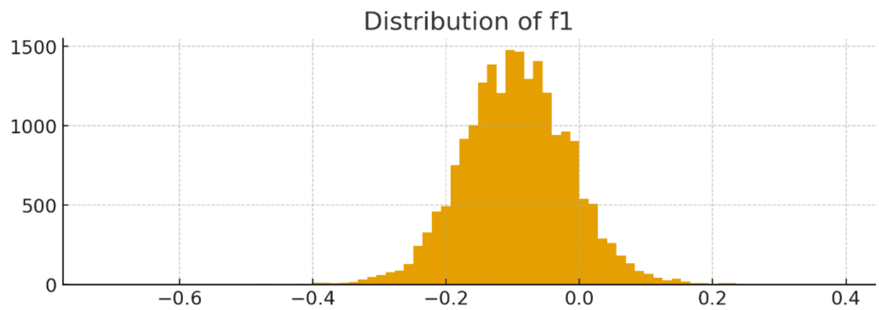
- **Add temporal context:** Introduce timestamps to use sequence models (LSTM Autoencoder or Temporal Convolutional Networks).
- **Hybrid ensemble:** Combine Isolation Forest and Autoencoder scores for more robust results.
- **Domain-informed labelling:** Collaborate with experts to validate anomaly instances and retrain models.
- **Adaptive thresholding:** Use dynamic or probabilistic thresholds instead of fixed percentiles.
- **Visualization dashboard:** Build an interactive tool for anomaly review and feedback loops.

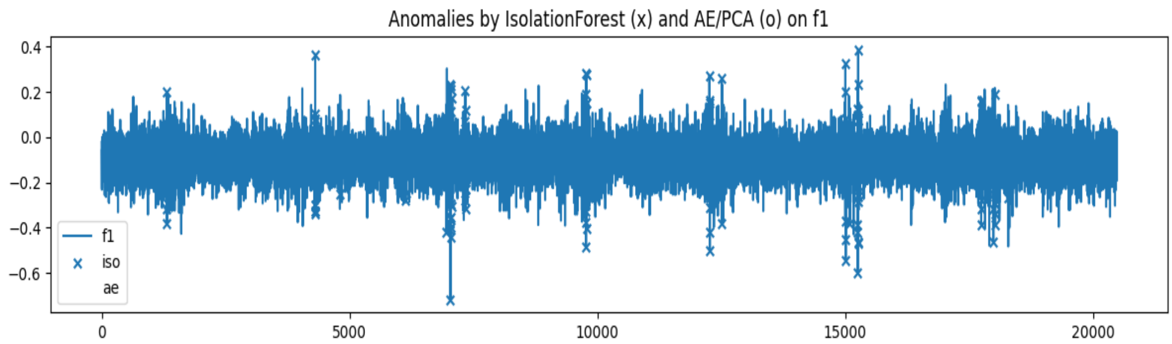
Conclusion

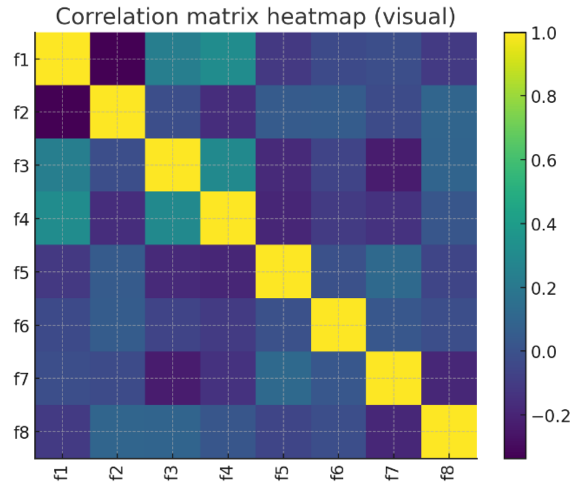
This project successfully demonstrates an **end-to-end unsupervised anomaly detection** using NASA sensor data.
The combination of **Isolation Forest** (statistical isolation) and **Autoencoder** (reconstruction-based) provides a robust foundation for detecting hidden anomalies in high-dimensional sensor systems. With future extensions to include temporal modelling and domain feedback, this approach can significantly enhance reliability and predictive maintenance in aerospace and industrial systems.

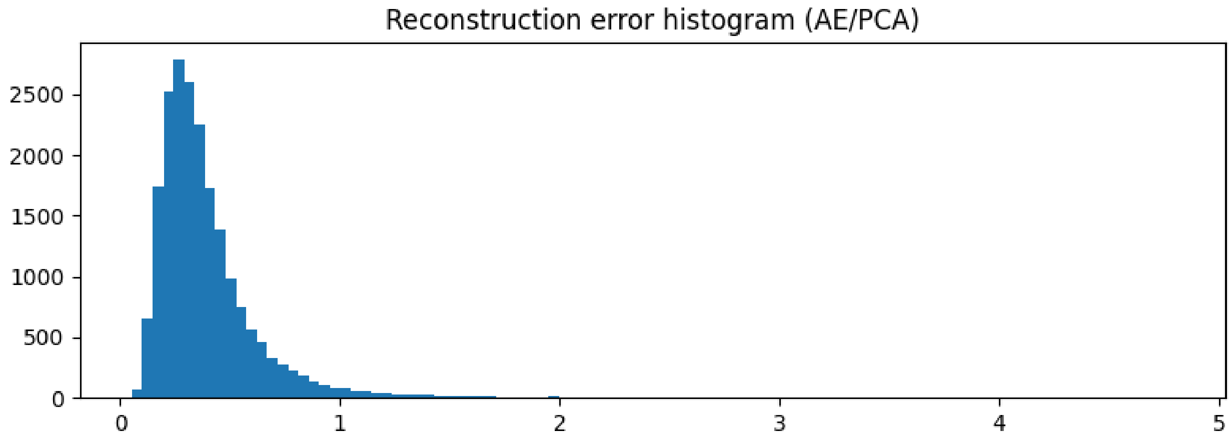
Below are some of the visualization results:








Correlation matrix heatmap (visual)
f1
f2
f3
f4
f5
f6
f7
f8
1.0
0.8
0.6
0.4
0.2
0.0
−0.2


Reconstruction error histogram (AE/PCA)
2500
2000
1500
1000
500
0
0
1
2
3
4
5